

Multi-Market Stock Volatility Forecasting Model

Nguyễn Thành Quy¹, Lê Trung Hoàng¹, and Trần Duy Hoàng¹

University of Science, Vietnam National University Ho Chi Minh City, Ho Chi Minh
City, Vietnam

lthoang@hcmus.edu.vn

Abstract. Forecasting daily stock-price volatility, how much a stock’s price is likely to swing on a given day, is a core input to risk management and option pricing: a risk manager sizing exposure or a trader pricing an option needs to know not only which direction a price may move but how large its swings will be. We propose a per-stock volatility forecasting model built on a gamma-loss gradient-boosting machine (GBM) over own-history features, and we ask where extra information actually helps. The own-history GBM already beats the HAR and HARQ linear benchmarks and a per-stock GARCH on QLIKE at every horizon. On the S&P 500, adding an External, forward-looking scheduled-earnings feature (GBME) cuts QLIKE by 8.7 to 11.2% over own-history at every horizon (Diebold-Mariano $p < 0.001$). A learned graph neural network (GNNHAR) does not beat the own-history GBM on either market. We replicate the study on a second, structurally different market, 405 Vietnam Ho Chi Minh Stock Exchange (HOSE) tickers, under an identical leakage-controlled walk-forward protocol scored on QLIKE with date-clustered Diebold-Mariano tests; the earnings lever does not transfer, which localizes the earnings gain to the S&P 500.

Keywords: Volatility forecasting · HAR · Gradient boosting · Earnings announcements · Diebold-Mariano.

1 Introduction

Financial markets are volatile: prices swing, and the size of those swings, an asset’s *volatility*, is the standard measure of its risk. Because volatility drives how options are priced, how much capital a position ties up, and how risk is budgeted across a portfolio, forecasting it accurately is a long-standing problem in financial econometrics [1]. Volatility is not observed directly; it is estimated from prices, so any forecast is a forecast of an estimate. We target the daily *Parkinson estimator* [2], which gauges a day’s variance from its high-low price range and uses more of the day’s information than a close-to-close measure; our forecasting target is therefore a variance (a squared quantity), not a standard deviation.

The standard linear approach is the HAR family [1]: a plain linear regression that predicts tomorrow’s volatility from its own averages over the last day, the

last week (about five trading days), and the last month (about twenty-two), summarising volatility’s long memory with a handful of coefficients. This benchmark is deliberately simple yet hard to beat. We build a stronger per-stock gradient-boosting model on top of it and ask what extra information actually helps: a forward-looking, per-firm earnings signal. As a learned-model reference we also train a graph neural network on the HAR features [10] and measure it under the same protocol.

We forecast daily Parkinson variance for the S&P 500 with a per-stock gamma-loss gradient-boosting model over own-history features, then add a forward-looking earnings feature and measure it under a leakage-controlled walk-forward protocol scored on QLIKE with date-clustered Diebold-Mariano tests. The lever that works is forward-looking and specific to the firm. A scheduled earnings release is known weeks ahead and raises volatility around the announcement [9]; encoding the distance from the forecast target to the nearest scheduled release cuts QLIKE by 8.7 to 11.2% over own-history at every horizon. A learned graph neural network (GNNHAR), by contrast, does not beat the own-history GBM on either market.

This paper contributes two results. First, a per-stock gamma-GBM over own-history features is a strong, transparent baseline for daily volatility that beats HAR, HARQ, a per-stock GARCH, and a learned graph neural network (GNNHAR) at every horizon. Second, a forward-looking scheduled-earnings feature is the dominant lever on the S&P 500, cutting QLIKE by 8.7 to 11.2% at every horizon (Diebold-Mariano $p < 0.001$); we replicate the study on a second market, 405 Vietnam HOSE tickers, where the earnings lever does not transfer, which localizes the earnings gain to the S&P 500.

Terminology. *Volatility*: the size of a stock’s price fluctuations, the standard proxy for its risk. *Parkinson estimator*: a day’s variance measured from its high-low price range; our forecasting target is this variance (squared units), not the standard deviation. *HAR*: a linear model predicting volatility from its own daily, weekly, and monthly averages. *QLIKE*: a proxy-robust error measure for volatility forecasts that stays reliable when the target is a noisy proxy [4]. *Diebold-Mariano (DM) test*: a test of whether one forecast is more accurate than another by more than chance, computed from their day-by-day loss differences [5]. *Horizon h* : how many trading days ahead we forecast ($h \in \{1, 5, 10, 22\}$, roughly a day, week, fortnight, and month).

2 Method and Setup

Target and metric. We forecast daily Parkinson variance $pk_t = \ln(H_t/L_t)^2/(4 \ln 2)$ from the daily high H and low L ($\ln$ is the natural logarithm), at horizons $h \in \{1, 5, 10, 22\}$ over 497 S&P 500 constituents. We report QLIKE as the primary loss [4], alongside MSE, RMSE and MAE. We test significance with a Diebold-Mariano statistic [5] on per-trading-date mean loss, with the Harvey-Leybourne-Newbold small-sample correction [6], because within-day stock errors are dependent.

Model. The forecasting model is a per-stock GBM (a gamma-loss Gradient Boosting Machine [7]: scikit-learn HistGradientBoostingRegressor [8], gamma-deviance loss equal to QLIKE up to a constant, 300 iterations, learning rate 0.05, 31 leaves, L2=1.0, 3-seed ensemble). Its own-history feature vector is $\mathbb{R}^8$: three HAR lags [1] and five log-volatility momentum terms. An optional External block (GBME) extends it: $\mathbb{R}^4$ of forward-looking scheduled-earnings features, chiefly the distance from the forecast target to the nearest scheduled release. Figure 2 shows the architecture.

Baselines. HAR [1] fits three own-history lags by ordinary least squares. HARQ adds a realized-quarticity interaction [3]. GARCH is a per-stock GARCH(1,1) on daily returns. GNNHAR [10] is a learned graph neural network (two graph-convolution layers over the three HAR lags, its best variant). All share the same target, protocol, and positivity floor as the GBM.

Features

Own-history block (8 features, used by GBM). Every own-history feature is observable at the forecast origin t . The three HAR lags capture short, medium, and long memory of realized variance over one day, one trading week (WK = 5), and one trading month (22 days):

$$\text{har_daily} = \text{pk}_t, \quad \text{har_weekly} = \frac{1}{5} \sum_{k=0}^4 \text{pk}_{t-k}, \quad \text{har_monthly} = \frac{1}{22} \sum_{k=0}^{21} \text{pk}_{t-k}. \quad (1)$$

Five log-volatility momentum terms, built on $\ell_t = \ln(\max(\text{pk}_t, \varepsilon))$ with $\varepsilon = 10^{-8}$, encode recent trend, multi-scale slope, deviation from the recent mean, and a standardized 22-day level:

$$\text{mr_change} = \ell_t - \ell_{t-1}, \quad \text{mr_slope5} = \frac{\ell_t - \ell_{t-5}}{5}, \quad \text{mr_slope10} = \frac{\ell_t - \ell_{t-10}}{10}, \quad (2)$$

$$\text{mr_dev5} = \ell_t - \frac{1}{5} \sum_{k=0}^4 \ell_{t-k}, \quad \text{mr_z22} = \frac{\ell_t - \mu_{22}(\ell)}{\sigma_{22}(\ell) + \varepsilon}, \quad (3)$$

where μ_{22} and σ_{22} are the trailing 22-day mean and standard deviation of ℓ . Together these summarize the stock’s own short-run acceleration and regime level.

External earnings block (4 features, used by GBME). Each earnings feature is evaluated at the target day $t + h$ from scheduled release dates known weeks ahead, so the block is forward-looking and leakage-safe. Let nxt be the trading days from $t + h$ to the next scheduled announcement, prv the trading

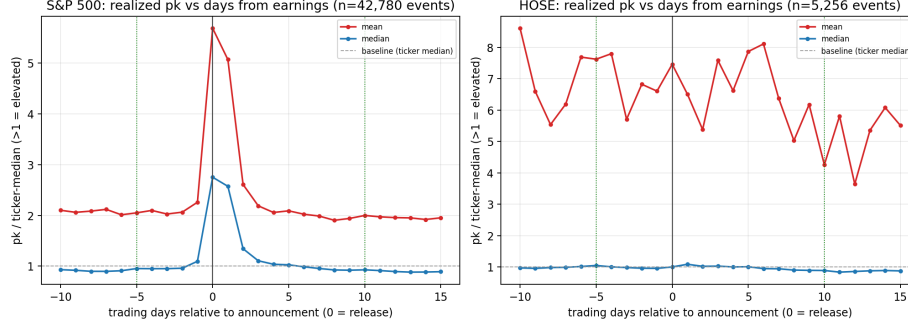


Fig. 1. Event study: median realized Parkinson variance (normalised by each ticker’s median) versus trading days from a scheduled earnings release. The S&P 500 median rises to about 2.75 times baseline on the announcement day and returns within roughly a week; the HOSE median stays near one at every offset, showing no announcement spike, which is why the earnings feature helps the S&P 500 but not HOSE.

days since the previous one, and $\text{dist} = \min(\text{nxt}, \text{prv})$:

$$\text{earn_prox} = \max\left(0, 1 - \frac{\text{dist}}{5}\right), \quad \text{earn_soon} = \mathbf{1}[\text{dist} \leq 3], \quad (4)$$

$$\text{earn_pre} = \max\left(0, 1 - \frac{\text{nxt}}{5}\right), \quad \text{earn_post} = \max\left(0, 1 - \frac{\text{prv}}{10}\right). \quad (5)$$

Here earn_prox is a symmetric triangular proximity that peaks at 1 on the announcement day and reaches 0 once the nearest release is at least a week away, earn_soon is a sharp ± 3 -day flag, earn_pre rises into the next release, and earn_post decays over a longer 10-day window because post-earnings volatility drift persists longer than the pre-announcement run-up. Away from any release all four are zero, so GBME reduces to GBM on ordinary days and activates only in the roughly two-week window around a scheduled release. Figure 1 shows the S&P 500 median variance rises to about 2.75 times baseline on the announcement day, which motivates these windows; on HOSE the median stays flat, so the block is inert there.

Protocol. We use expanding-window walk-forward folds with a target-horizon embargo. All correlations, adjacencies, factor estimates, and model fits use training data only; the test window is read once. Every gradient-boosting model averages three seeds. Predictions are floored identically across models before scoring.

3 Results: The Own-History GBM Beats the Baselines

Table 1 reports all four metrics for the S&P 500 across GARCH, HAR, HARQ, the learned GNNHAR, the own-history GBM, and the earnings model GBME at every horizon. The own-history GBM beats HAR and HARQ on QLIKE at every horizon, and the learned GNNHAR does not beat it: GNNHAR is worse

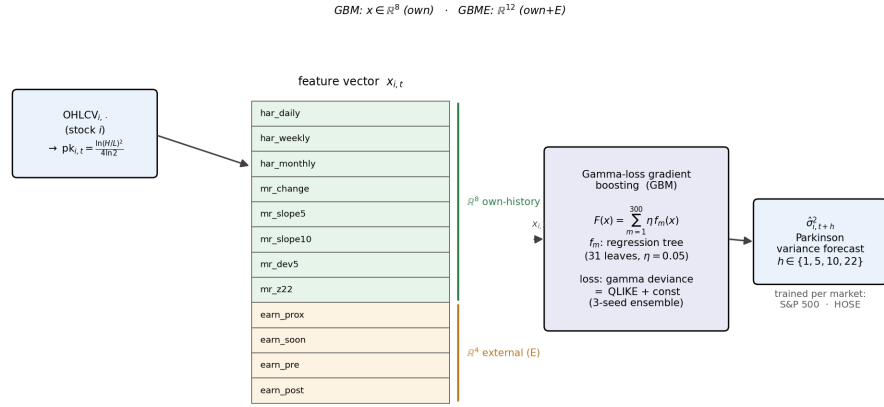


Fig. 2. Model architecture. A per-stock feature vector combines own-history features ($\mathbb{R}^8$: three HAR lags and five log-volatility momentum terms) with an optional External forward-looking scheduled-earnings block ($\mathbb{R}^4$). A gamma-loss gradient-boosting ensemble maps the feature vector to the multi-horizon Parkinson-variance forecast.

than the GBM at every horizon (QLIKE worse by 5.0 to 10.6%, DM $p < 10^{-10}$) and worse than even linear HAR at the three shortest horizons, so a multi-hop graph network recovers nothing the own-history GBM does not already capture. The earnings model GBME carries the bolded QLIKE in every block; the next section isolates why.

Table 1. Main result, S&P 500 (497 constituents; walk-forward, 3-seed ensembles). Lower is better on every metric. $\text{MSE} \times 10^{-6}$, $\text{RMSE/MAE} \times 10^{-3}$; QLIKE unscaled. GARCH is a per-stock GARCH(1,1) benchmark (QLIKE only); it is omitted (-) on the S&P 500 owing to an unstable per-stock fit. Bold marks the best value in each metric column per horizon; the QLIKE bold additionally requires a significant date-clustered Diebold-Mariano test ($p < 0.05$).

h	Model	MSE	RMSE	MAE	QLIKE
1	GARCH	–	–	–	–
1	HAR	0.575	0.759	0.213	0.370
1	HARQ	0.565	0.752	0.215	0.378
1	GNNHAR	0.576	0.759	0.212	0.375
1	GBM	0.557	0.746	0.209	0.357
1	GBME	0.540	0.735	0.202	0.317
5	GARCH	–	–	–	–
5	HAR	0.614	0.784	0.230	0.436
5	HARQ	0.613	0.783	0.231	0.438
5	GNNHAR	0.638	0.799	0.233	0.446
5	GBM	0.611	0.782	0.226	0.412
5	GBME	0.596	0.772	0.218	0.371
10	GARCH	–	–	–	–
10	HAR	0.621	0.788	0.236	0.461
10	HARQ	0.621	0.788	0.236	0.461
10	GNNHAR	0.660	0.812	0.243	0.477
10	GBM	0.629	0.793	0.232	0.431
10	GBME	0.615	0.784	0.225	0.392
22	GARCH	–	–	–	–
22	HAR	0.633	0.795	0.242	0.485
22	HARQ	0.631	0.794	0.241	0.483
22	GNNHAR	0.635	0.797	0.241	0.476
22	GBM	0.623	0.790	0.236	0.447
22	GBME	0.609	0.781	0.228	0.408

4 Forward-Looking Earnings Are the Lever

A diagnostic locates the missing lever. The own-history GBM beats HAR across the calm deciles, yet its remaining QLIKE concentrates in the storm decile: decile D9 alone carries 43% of the model’s total test QLIKE (Fig. 3). The model under-forecasts exactly there, where its median forecast-to-realized ratio falls below one while every calm decile sits above one (Fig. 4). Permutation importance ranks the three own-history HAR lags far above every momentum and auxiliary term (Fig. 5), so own-history is exhausted and the storm component needs a signal from outside the stock’s past. A scheduled earnings release is known weeks ahead and drives the announcement-driven part of these spikes, which makes a forward-looking earnings feature the natural lever for the residual the GBM cannot reach.

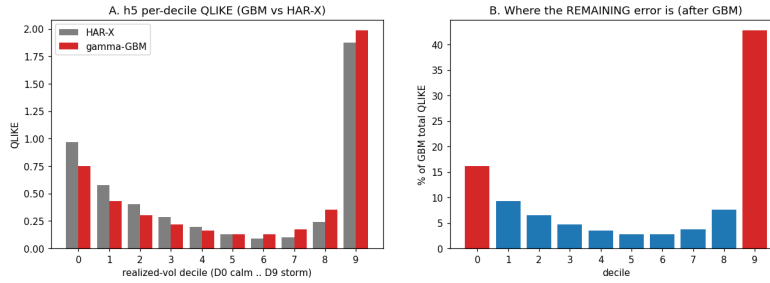


Fig. 3. The own-history GBM beats HAR on the calm deciles, and its remaining QLIKE concentrates in the storm decile D9, which carries 43% of the model total.

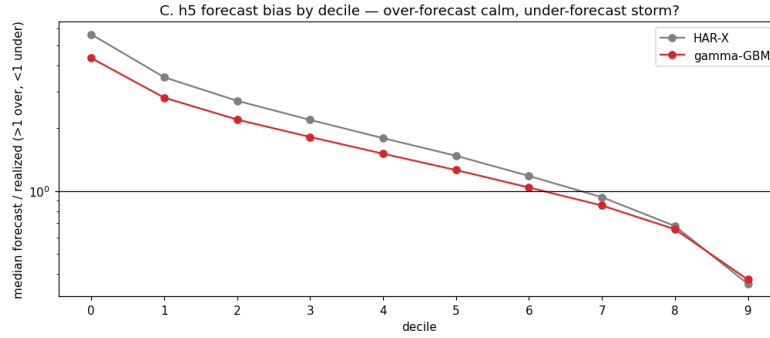


Fig. 4. The GBM under-forecasts the storm deciles, where its median forecast-to-realized ratio drops below one.

The earnings feature is the lever. In Table 1, the External earnings model GBME posts the lowest QLIKE at every horizon. The QLIKE gain over own-history is 8.7 to 11.2% and is significant at every horizon (DM GBME vs GBM $p < 0.001$ at $h1$, $h5$, $h10$, $h22$), which is why GBME carries the bolded QLIKE in every block. Against the HAR baseline the earnings model gains 14 to 16%. Nonlinearity and the QLIKE-loss alignment, isolated by the own-history GBM over HAR, contribute a further 3.4%.

Earnings do not fix the storm tail. Decomposed by realized-volatility decile, the model's error still concentrates in the top decile, which carries roughly ten times the calm-day QLIKE, and its forecast-to-realized ratio there is about 0.4. The model compresses its predictions toward the conditional mean because the storm magnitude is absent from origin-time features. The residual is exogenous, so no origin-time feature reaches it.

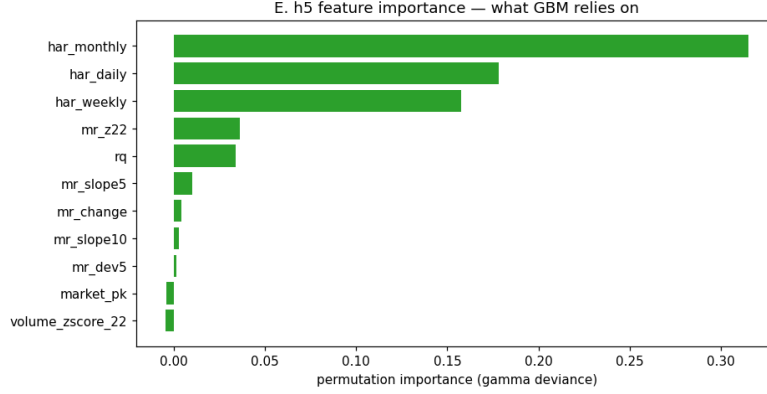


Fig. 5. Permutation importance ranks the own-history HAR lags far above every other feature, so own-history is exhausted.

5 Related Work

Neural and graph volatility forecasting. A recent literature forecasts volatility with graph neural networks that wire stocks through correlation or spillover edges [10]. We include the published GNNHAR as a learned baseline and find, under a leakage-controlled QLIKE walk-forward, that it does not beat a per-stock gradient-boosting model over own-history features. Prior Vietnamese work builds correlation and centrality networks to predict index volatility at a monthly horizon without a HAR or QLIKE benchmark; we extend that setting to daily per-firm QLIKE on 405 HOSE tickers.

HAR and its extensions. HAR [1] and HARQ [3] model volatility from own history and a realized-quarticity interaction. We keep HAR and HARQ as baselines and show that a QLIKE-loss gradient-boosting model [7, 8] with an External earnings feature dominates them, while QLIKE [4] and the Diebold-Mariano test [5, 6] provide the proxy-robust evaluation.

6 Multi-Market Replication: Vietnam HOSE

We replicate the study on a second, structurally different market. On 405 Ho Chi Minh Stock Exchange (HOSE) tickers under the identical protocol, forecasting daily Parkinson variance at the same four horizons with 390,577 to 399,040 stock-day observations per horizon, the own-history GBM again wins and the earnings lever does not transfer.

The own-history GBM wins on HOSE too. Table 2 reports all four metrics. The own-history GBM is the DM-verified best model at every horizon: no feature block significantly lowers its QLIKE, while HAR and HARQ trail by a wide margin. The External earnings feature does not improve it: GBME is statistically indistinguishable from GBM at every horizon (DM GBME vs GBM

Table 2. Multi-market replication, HOSE (405 tickers; 390,577 to 399,040 stock-day observations per horizon). Lower is better on every metric. $\text{MSE} \times 10^{-6}$, $\text{RMSE}/\text{MAE} \times 10^{-3}$; QLIKE unscaled. GNNHAR is reported at $h1$ and $h5$. GARCH is a per-stock GARCH(1,1) benchmark scored on QLIKE over the estimable subset ($\sim 98.6\%$ of rows); its squared-error cells are not comparable and shown as -. Bold marks the best value in each metric column per horizon; the QLIKE bold additionally requires a significant date-clustered Diebold-Mariano test ($p < 0.05$).

h	Model	MSE	RMSE	MAE	QLIKE
1	GARCH	-	-	-	1.654
1	HAR	0.660	0.813	0.519	1.812
1	HARQ	0.639	0.799	0.506	1.785
1	GNNHAR	0.585	0.765	0.438	1.682
1	GBM	0.541	0.735	0.394	1.570
1	GBME	0.545	0.739	0.396	1.569
5	GARCH	-	-	-	1.740
5	HAR	0.663	0.814	0.523	1.822
5	HARQ	0.652	0.807	0.515	1.807
5	GNNHAR	0.616	0.785	0.469	1.744
5	GBM	0.579	0.761	0.427	1.648
5	GBME	0.580	0.761	0.428	1.649
10	GARCH	-	-	-	1.780
10	HAR	0.666	0.816	0.525	1.828
10	HARQ	0.659	0.812	0.520	1.818
10	GBM	0.596	0.772	0.442	1.684
10	GBME	0.596	0.772	0.443	1.684
22	GARCH	-	-	-	1.838
22	HAR	0.672	0.820	0.528	1.835
22	HARQ	0.668	0.817	0.525	1.830
22	GBM	0.621	0.788	0.462	1.726
22	GBME	0.623	0.789	0.462	1.726

$p = 0.79, 0.37, 0.79, 0.88$ at $h1, h5, h10, h22$). The learned GNNHAR again loses to the GBM (at $h1$ and $h5$, QLIKE worse by 5.8 to 7.1%, DM $p < 10^{-80}$). The own-history GBM therefore carries the bolded QLIKE in every block.

Earnings does not transfer to Vietnam. Using real crawled Vietnam announcement dates, GBME is statistically indistinguishable from GBM at every horizon. A per-stock, per-day diagnostic on the HOSE test set locates two compounding causes. First, the feature fires on only 5.6% of test observations, and 0.84% before 2025, so it cannot move the pooled metric across most of the panel. Second, where it fires the volatility response is muted: realized variance on announcement-proximate days runs about 1.1 times the baseline, against the pronounced S&P 500 earnings spike, because Vietnam’s daily price limits cap the informational jump. The no-transfer result holds in the well-covered 2025-2026 folds, so it does not rest on missing dates in that window.

The HOSE result is robust to regime spikes. The own-history GBM beats HAR in every walk-forward fold (per-fold QLIKE 1.37 to 1.83 against HAR’s 1.81 to 1.84), and the margin survives excluding the COVID (February to April 2020), 2022, and April 2025 tariff shock windows: dropping those 57,466 observations (14% of the panel) leaves the $h1$ QLIKE gain over HAR at 13.8% (from 13.3% on the full panel) and keeps every horizon significant (DM $p < 10^{-25}$). The GBM advantage is a calm-market property, not an artifact of a few crisis days.

Takeaway. The own-history GBM wins on a second, structurally different market. The earnings lever is specific to the S&P 500 and does not generalize to HOSE under the available real announcement dates.

7 Limitations

Earnings dates. The earnings feature measures distance to the nearest scheduled release from an archive that records realized dates, not a point-in-time scheduling snapshot. A date-jitter check shows the gain does not depend on exact-date timing: perturbing every S&P 500 earnings date by a uniform random offset, the $h1$ QLIKE gain over own-history falls from 11.1% with true dates to 9.0% at plus or minus one trading day, 6.6% at plus or minus three days, and 4.5% at plus or minus five days, and stays positive at every horizon. A point-in-time scheduling calendar remains to be added to control rescheduling within the forecast window.

Vietnam coverage. The Vietnam earnings test draws on real announcement dates that are comprehensive only for 2025-01 through 2026-09, plus a sparse 2016-2026 news feed covering roughly 218 tickers. We therefore claim no incremental earnings value on HOSE under the available real dates, rather than a definitive full-history null.

Scope. The analysis is conditional on the Parkinson daily-OHLC regime and does not speak to intraday or option-implied signals.

8 Conclusion

We built a per-stock volatility forecasting model on a gamma-loss gradient-boosting machine over own-history features and asked where extra information helps. The own-history GBM beats the HAR, HARQ, and GARCH benchmarks and a learned graph neural network (GNNHAR) at every horizon. On the S&P 500, an External, forward-looking scheduled-earnings feature cuts QLIKE by 8.7 to 11.2% at every horizon (Diebold-Mariano $p < 0.001$). We replicated the study on 405 Vietnam HOSE tickers under an identical protocol: the own-history GBM again wins and the earnings lever does not transfer, which localizes the earnings gain to the S&P 500. The practical takeaway is that, for daily volatility, a strong own-history model plus a forward-looking per-firm signal is what matters, and the value of that signal depends on the market’s information and microstructure.

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